MIL-E-1/1301D(EC)
27 June 1983
SUPERSEDING
MIL-E-1/1301C(NAVY)
22 January 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 7308

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Twin triode, medium Mu

Outline -- 6-2 (EIA)

Base -- E9-1

Envelope -- T-6 1/2

Cathode -- Coated unipotential

Base connections:

Pin No.	--	1	2	3	4	5	6	7	8	9
Element	--	2a	2g	2k	h	h	1a	1g	1k	sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ef V	Ebb V dc	Eb V dc	Ecc V dc	Ec V dc	Ehk v	Rk/k ohms	Rg/g Meg
Maximum:	6.6	---	250	---	---	70 to -135	---	0.5 (See note 1)
Minimum:	6.0	---	---	---	-110	---	---	---
TEST CONDITIONS:	6.3	100	90 (approx.)	+9	---	---	680	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ik/k mA dc	Ic/g mA dc	Pp/p W	TE °C	Alt ft
Maximum:	22	---	1.65	165	Note 2
Minimum:	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.65	II	If	305	365	mA
1336	Heater-cathode leakage			0.65	II	Ihk	---	10	µA dc
1266	Total grid current (1)		Ebb = 91 V dc; Ecc/g = -1.5 V dc; Rk/k = 0	0.65	II	Ic	0	-0.1	µA dc
1256	Electrode current (1) (anode)	3	Ebb = 90 V dc; Ecc = 0; Rk = 80 ohms	0.65	II	Ib	11.3	18.7	mA dc
1256	Electrode current (2) (anode)	3	Ec = -15 V dc; Eb = 150 V dc	0.65	II	Ib	---	5	µA dc
1306	Transconductance (1)	3		0.65	II	Sm	10,400	14,600	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	3	g-all p-all 10 meg in series	2.5	S3	R	100	---	Meg
1306	Transconductance (2)	3	Ef = 5.7 V ac	2.5	I	$\frac{\Delta Sm}{Ef}$	---	15	%
1316	Amplification factor			6.5	I	Mu	26.5	39.5	---
1266	Total grid current (2)	3,4		2.5	I	Ic	0	-0.5	µA dc
2201	Noise and microphonics	3	Ebb = 250 V dc; Ecal = 10 mV ac; Ck = 100 µF; Rk = 680 ohms; Rp = 10,000 ohms	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance		No shield						
		3				Cgp	1.2	1.6	pF
		3				Cin	2.7	3.9	pF
			Section 1	6.5	Code E	Cout	1.6	2.0	pF
			Section 2			Cout	1.5	1.9	pF
						Cgg	---	.008	pF
						Cpp	---	.060	pF

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 -Continued								
1031	Low-frequency vibration	5	Rp = 2,000 ohms; Ck = 1,000 µF	6.5	Code H	Ep	---	50	mV ac
1041	Shock	6	G = 450; Ehk = +100 V dc	6.5	See note 7	---	---	---	---
1031	Vibration-fatigue		F = 50	6.5	See note 7	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	Low-frequency vibration	5	Rp = 2,000 ohms; Ck = 1,000 µF	---	---	Ep	---	75	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	15	µA dc
1306	Transconductance (1)	3		---	---	Sm	9,000	16,500	µmhos
1266	Total grid current (1)		Ebb = 91 V dc; Ecc/c = -1.5 V dc; Rk/k = 0	---	---	Ic	0	-0.2	µA dc
1211	Base strain			---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life	8	Ef = 7.5 V; Ehk = +135 V dc; Ec = Eb = 0; 1 minute on, 4 minutes off	---	---	---	---
---	Heater-cycling life-test end point						
1336	Heater-cathode leakage			Ihk	---	20	µA dc

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		t
					Min	Max	
	Quality conformance inspection, part 3 - Continued						
1516	Stability life	3,8	Ehk = +135 V dc; Rg = 47,000 ohms; TA = Room	---	---	---	---
---	Stability life-test end point						
1306	Change in transconductance (1) of individual tubes	3		ΔSm t	---	10	%
1501	Intermittent life	3,9	Ehk = +135 V dc; Rg = 47,000 ohms; TE = 165°C (min)	---	---	---	---
---	Intermittent life-test end points (500 hours)	10					
1266	Total grid current (1)		Ebb = 91 V dc; Ecc/c = -1.5 Vdc; Rk/k = 0	Ic	0	-0.9	μA d
1301	Heater current			If	305	365	mA
1306	Change in transconductance (1) of individual tubes	3		ΔSm t	---	15	%
1306	Transconductance (2)	3	Ef = 5.7 V ac	ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	20	μ c
1211	Insulation of electrodes	3	g - all p - all 10 Meg in series	R	50	---	Meg
1306	Average change in transconductance (1)	3		Avg ΔSm t	---	15	%
---	Intermittent life-test end points (1,000 hours)	11					
1266	Total grid current (1)		Ebb = 91 V dc; Ecc/c = -1.5 Vdc; Rk/k = 0	Ic	0	-0.9	μA
1301	Heater current			If	305	365	mA
1306	Change in transconductance (1) of individual tubes	3		ΔSm t	---	25	%
1336	Heater-cathode leakage			Ihk	---	20	μA
1211	Insulation of electrodes	3	g - all p - all 10 Meg in series	R	50	---	Me

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NOTES:

1. This value is for operation under fixed bias conditions. With cathode bias, Rg may be 1 Meg maximum.

2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

3. Test each unit separately.

4. Prior to this test, tubes shall be preheated 5 minutes with both sections operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Total grid current (2) shall be the last test performed on the sample selected for the total grid current (2) test.

Ef	Ecc	Ebb	Rk	Rg
V	V dc	V dc	Ohms	Ohms
7	+9	100	680	47,000

5. Tie 1k to 2k; 1g to 2g; and 1a to 2a. Parasitic suppressors of 50 ohms shall be permitted.

6. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.

7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of a lot failure, the lot shall be rejected, and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

8. The AQL (percent defective) shall be 1.0, inspection level S3.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

10. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

11. The allowable defectives per characteristic shall not be greater than 2 for the first sample and shall not be greater than 5 for combined samples.

12. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
Navy - SH, AS, OS, MC, CG

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-N189)

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